

IDS568 Milestone 5: Governance Memo

Overview

This system implements caching and batching to improve large language model inference performance. While these optimizations improve latency and throughput, they introduce governance considerations related to privacy, data retention, and system misuse.

Privacy Considerations

Caching stores prompts and responses to improve performance. This introduces privacy risks if sensitive user inputs are stored in memory.

To mitigate this risk:

- Cache keys are hashed
- No plaintext user identifiers are stored
- Cache entries expire using TTL

These steps reduce the risk of exposing sensitive information.

Data Retention Policy

Cached responses are stored temporarily using a configurable TTL. This ensures:

- Old responses expire automatically
- Sensitive prompts are not retained indefinitely
- Memory usage remains controlled

Short TTL values reduce privacy risks while maintaining performance benefits.

Potential Misuse Scenarios

Potential misuse includes:

- Caching sensitive prompts
- Returning stale responses
- High request load causing degraded performance

These risks are mitigated through:

- Cache expiration
 - Concurrent request handling
 - Dynamic batching controls
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Compliance Considerations

Caching introduces potential data residency and privacy compliance concerns.

Mitigation strategies include:

- Avoid storing user identifiers
- Use short retention windows
- Restrict cached data to model inputs only

These approaches align with privacy-aware system design.

Implementation-Specific Governance Controls

This system implements several governance safeguards directly in the implementation. Cache keys are generated using hashed prompt inputs, ensuring that no plaintext user identifiers are stored in memory. The caching layer uses a configurable time-to-live (TTL) value, which defaults to 300 seconds, ensuring that responses expire automatically and reducing long-term retention risks.

Additionally, the cache includes a maximum entry limit of 1000 responses, which prevents unbounded memory growth and reduces exposure of stored data. Expired entries are removed automatically, ensuring that stale responses are not returned indefinitely.

These safeguards help maintain privacy-aware inference behavior while still enabling performance optimizations through batching and caching.

Threat Model and Risk Analysis

Caching introduces several governance risks beyond basic privacy concerns. If sensitive prompts are stored, an attacker with access to system memory, logs, or cache contents could potentially recover confidential inputs or outputs. In addition, malicious prompt injection content could persist in cache and be returned repeatedly for later requests, extending the lifetime of harmful or manipulated outputs.

There is also a retention trade-off. Longer cache lifetimes improve reuse and performance, but they increase the period during which stored outputs remain exposed. For that reason, short TTL values are safer in privacy-sensitive environments. In this implementation,

automatic expiration reduces long-term retention risk and helps prevent stale responses from remaining indefinitely.

In regulated environments, retention and minimization policies matter. For example, GDPR-style data minimization principles would favor shorter retention windows and stricter controls over stored content. In a production deployment, stronger safeguards such as encrypted caches, restricted access controls, user opt-out policies, and deletion workflows would further reduce governance risk.

Conclusion

Batching and caching improve performance but introduce governance risks. The implemented safeguards ensure the system remains privacy-aware while maintaining performance improvements.
